# Class BankAccount

java.lang.Object
    BankAccount

public class **BankAccount**
extends Object

The BankAccount class works to represent a savings bank account, tracking aspects such as balance, total deposits / withdrawls, and interest earned. This also includes functions which add or subtract from such statistics, as well as calculating interest.

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## *Constructor Summary*

**Constructors**

| Constructor | Description |
|---|---|
| **BankAccount**() | Creates an empty bank accont, setting the monthly interest as an annual 4.4% divided into a monthly rate. |
| **BankAccount**(double inBalance, double intRate) | Creates a bank accont with desired starting balance and interest, which is divided into a monthly amount. |
| **BankAccount**(int inBalance, double intRate) | Creates a bank accont with desired integer starting balance and interest rate, which is divided into a monthly amount. |

## *Method Summary*

**All Methods** | **Instance Methods** | **Concrete Methods**

| Modifier and Type | Method | Description |
|---|---|---|
| void | **calcInterest**() | Calculates and adds monthly interest to an account if they have a positive balance. |
| double | **getBalance**() | Returns the current account balance. |
| double | **getDeposit**() | Returns the total amount deposited. |
| double | **getInterest**() | Returns the total amount of interest earned. |
| double | **getWithdraw**() | Returns the total amount withdrawn. |
| void | **makeDeposit**(double add) | Adds and tracks a deposit to an account. |
| void | **makeDeposit**(int add) | Adds and tracks a whole-number deposit to an account. |
| void | **makeWithdraw**(double sub) | Subtracts and tracks a withdrawl from an account. |
| void | **makeWithdraw**(int sub) | Subtracts and tracks a whole-number withdrawl from an account. |

### Methods inherited from class java.lang.Object

clone, equals, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

## *Constructor Details*

### BankAccount [Show source in BlueJ]

```
public BankAccount()
```

Creates an empty bank accont, setting the monthly interest as an annual 4.4% divided into a monthly rate.

### BankAccount [Show source in BlueJ]

```
public BankAccount(int inBalance,
                   double intRate)
```

Creates a bank accont with desired integer starting balance and interest rate, which is divided into a monthly amount.

**Parameters:**

`inBalance` - Starting balance in whole dollars.

`intRate` - The annual interest rate in percentage.

### BankAccount [Show source in BlueJ]

```
public BankAccount(double inBalance,
                   double intRate)
```

Creates a bank accont with desired starting balance and interest, which is divided into a monthly amount.

**Parameters:**

`inBalance` - Starting balance.

`intRate` - The annual interest rate in percentage.

## *Method Details*

### makeDeposit [Show source in BlueJ]

```
public void makeDeposit(double add)
```

Adds and tracks a deposit to an account.

**Parameters:**

`add` - The amount added to an account.

### makeDeposit [Show source in BlueJ]

```
public void makeDeposit(int add)
```

Adds and tracks a whole-number deposit to an account.

**Parameters:**

`add` - The whole-number amount added to an account.

### makeWithdraw [Show source in BlueJ]

```
public void makeWithdraw(double sub)
```

Subtracts and tracks a withdrawl from an account.

**Parameters:**

`sub` - The amount withdrawn from an account.

### makeWithdraw [Show source in BlueJ]

```
public void makeWithdraw(int sub)
```

Subtracts and tracks a whole-number withdrawl from an account.

**Parameters:**

`sub` - The whole-number amount withdrawn from an account.

### calcInterest [Show source in BlueJ]

```
public void calcInterest()
```

Calculates and adds monthly interest to an account if they have a positive balance.

### getBalance [Show source in BlueJ]

```
public double getBalance()
```

Returns the current account balance.

**Returns:**

The current account balance

### getDeposit [Show source in BlueJ]

```
public double getDeposit()
```

Returns the total amount deposited.

**Returns:**

The total amount of deposits

### getWithdraw [Show source in BlueJ]

```
public double getWithdraw()
```

Returns the total amount withdrawn.

**Returns:**

The total amount withdrawn

## getInterest [Show source in BlueJ]

```
public double getInterest()
```

Returns the total amount of interest earned.

**Returns:**

The total interest